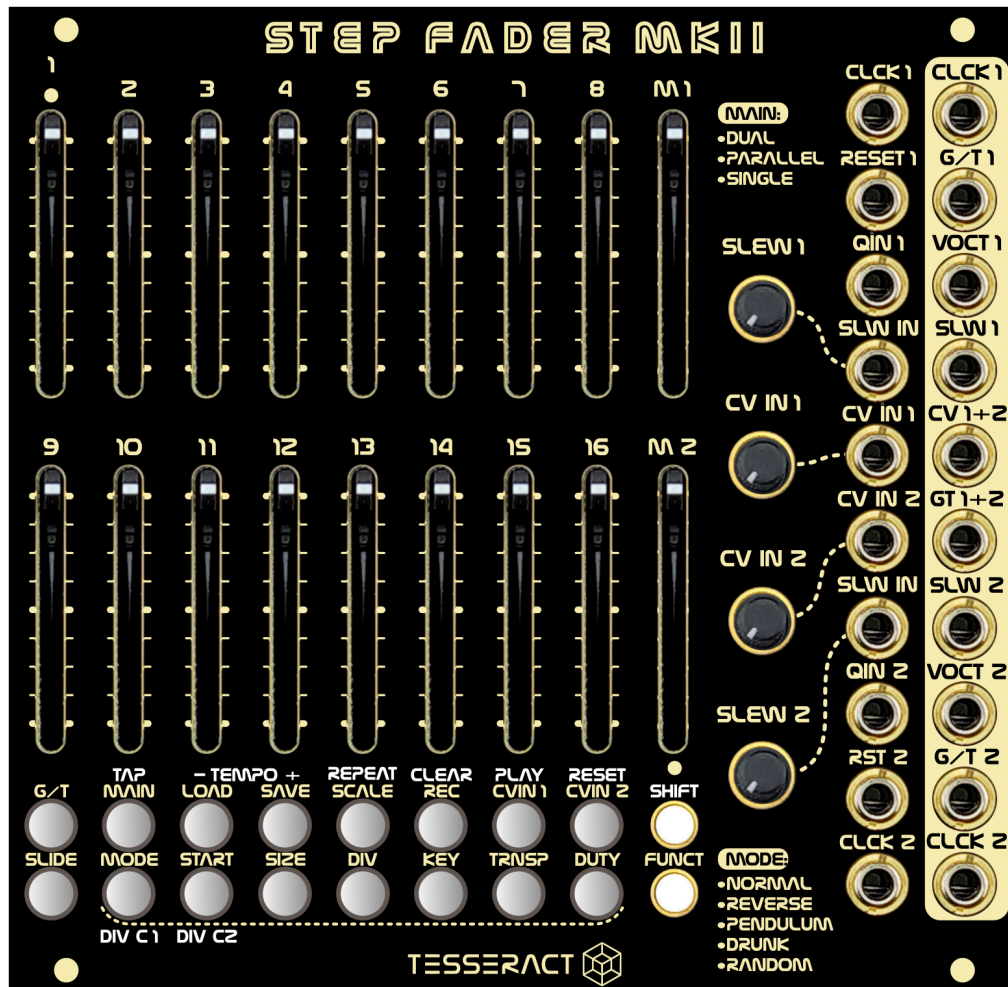


TESSERACT MODULAR



Step Fader mkII

Step Fader mkII it's a 'hands on' eurorack dual sequencer, featuring two 8 step lines or one with 16 steps plus quantizer, slew limiter, internal clock, CV inputs to control several parameters and more...

Each sequencer has independent settings for the play mode, start point, length, clock divider, transposition and duty.

The 2 channels of the quantizer can be used for processing external signals, and the same applies for the slew limiter circuit.

There are also 16 presets to store & recall all the sequencer settings.

The voltage range of the sequencers:

- First batch: 5v approx, and the sum output (CV 1+2) is about 10v.
- Rev 2.8: 9,7v approx, and the sum output (CV 1+2) is about 10,7v.

[ASSEMBLY GUIDE](#)

INPUTS



- **CLK 1** clock input for sequencer 1 (normalized to CLK 2)
- **RESET 1** reset input for sequencer 1
- **QIN 1** input for quantizer 1, range 0 to +10 volts
- **SLW IN** input for the slew limiter 1
- **CV IN 1** CV input for modulation
- **CV IN 2** CV input for modulation
- **SLEW IN** input for slew limiter 2
- **QIN 2** input for quantizer 2, range 0 to +10 volts
- **RESET 2** reset input for sequencer 2
- **CLK 2** clock input for sequencer 2

OUTPUTS



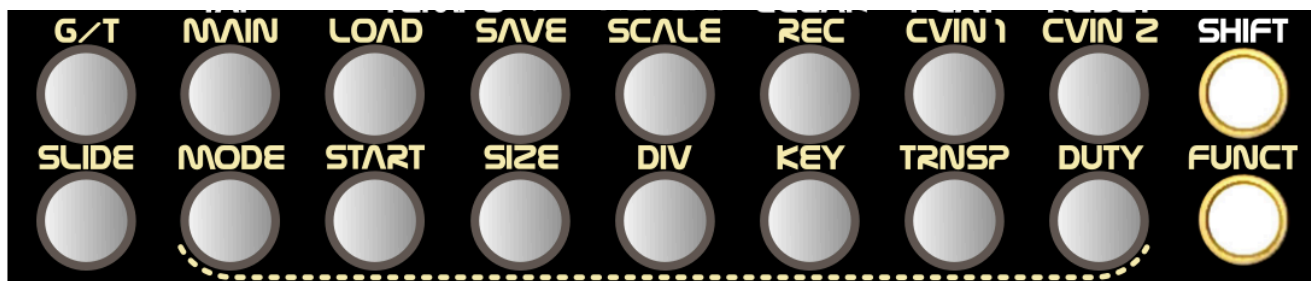
- **CLK 1** clock output for sequencer 1
- **G/T 1** gate & trigger 1 output
- **VOCT 1** quantizer 1 output (V/OCT)
- **SLW 1** slew limiter 1 output
- **CV 1 + 2** sum of both CVs (taken from both slew outs)
- **GT 1+2** sum of both G/T outputs (aka OR)
- **SLEW 2** slew limiter 2 output
- **VOCT 2** quantizer 2 output (V/OCT)
- **G/T 2** gate & trigger 2 output
- **CLK 2** clock output for sequencer 2

FUNCTION BUTTON AND KEYPAD MODES

The 16 illuminated buttons can be used to set several parameters, even to play the sequencer as a keyboard.

Click once the *FUNCT* button will enter the '*select keymode*', clicking again on one of the 16 illuminated buttons will select the overall behavior (aka *key mode*).

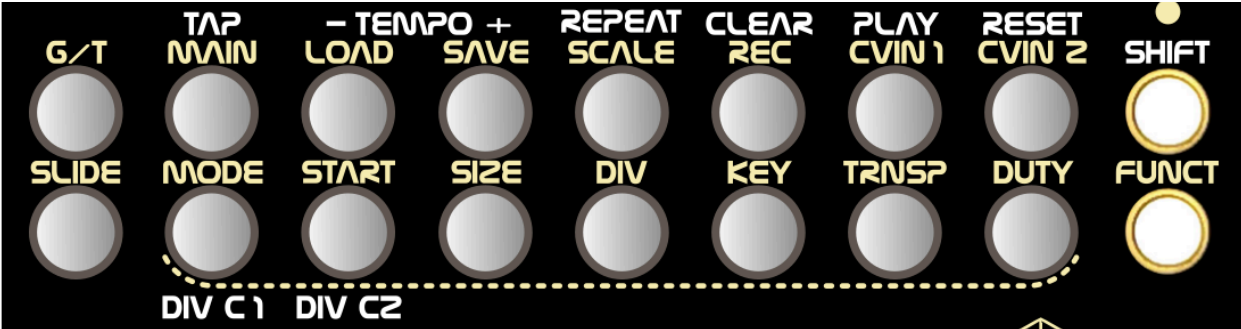
The selected *key mode* will remain active, with the exception of 'LOAD' and 'SAVE'. For those ones, the previous key mode will be recalled after loading or saving the actual state of the sequencer.



G/T	Gate / Trigger mode (long press will set gate, short set trigger)
MAIN	sets the 'Main Setup' of the module (dual, parallel, single)
LOAD	load preset
SAVE	save preset
SCALE	set the scale for the quantizer of both sequencers
REC	record mode
CV IN 1	set the destination for the CV input 1
CV IN 2	set the destination for the CV input 2
SLIDE	quantizer will do a slide when a step is active
MODE	set the play mode of each sequencer
START	set the starting step of each sequencer
SIZE	set the length of sequencer
DIV	set the clock divider of each sequencer
KEY	use the keypad to play like a sort of keyboard
TRNSP	set the transposition of each quantizer (from -19 to +19 semitones)
DUTY	set the duration of the triggers (from 5 milliseconds to $\frac{7}{8}$ of the step length)

SHIFT BUTTON SHORTCUTS

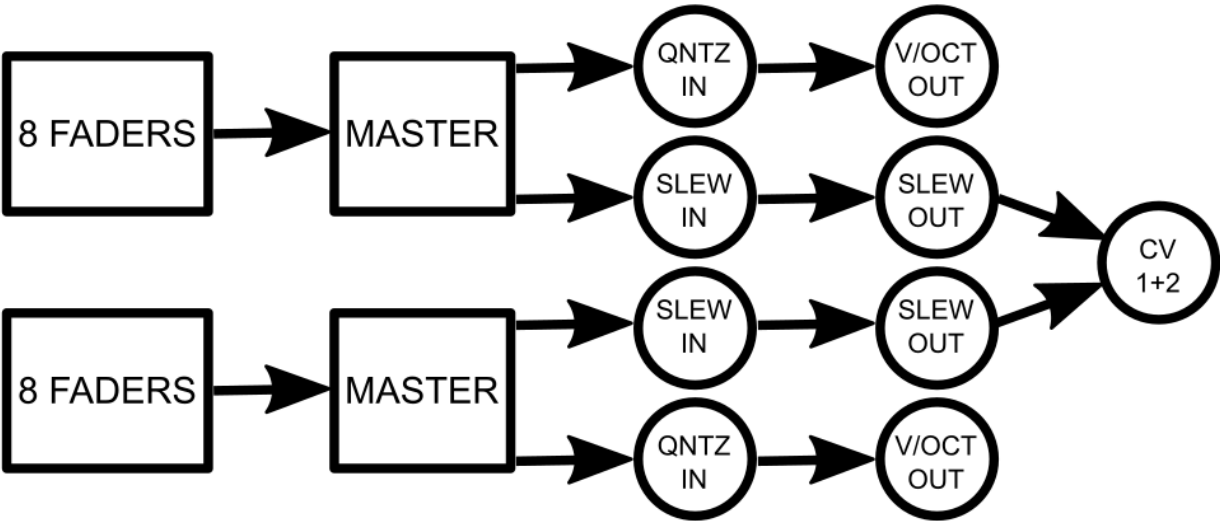
Several settings and actions can be done by holding the shift button while clicking on the illuminated buttons.



TAP	tap tempo for the internal clock generator
- TEMPO	rests 2 BPM to the internal clock
TEMPO +	adds 2 BPM to the internal clock
REPEAT	set on/off the step repeat / re-trigger (only affects if the clock divider is bigger that 1)
CLEAR	clears the recorded sequence
PLAY	play / stop for the internal clock
RESET	manual reset for both sequencers
DIV C1	sets on/off the clock division for the clock output 1
DIV C2	sets on/off the clock division for the clock output 2

SIGNAL FLOW

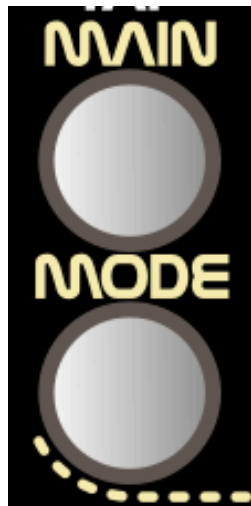
The microprocessor controls clocks, gates, quantizers and the position of the sequencers (it decides which step will be active). The rest of the signal flow is purely analog.



M1 and M2 are the master faders, you can think about them as ‘voltage range’ controls for each row.

To use the Step Fader mkII as a 16 steps sequencer (main mode = single) a cable from the CV 1+2 has to be connected to any of the quantizer inputs, if not, each quantizer will process their respective row of faders.

MAIN SETUP AND PLAY MODES



MAIN SETUP:

- DUAL 2 totally independent 8 step sequencers
- PARALLEL one 8 step sequencer with 2 rows of CV
- SINGLE one sequencer up to 16 steps

PLAY MODES:

- Normal
- Reverse
- Pendulum
- Drunk
- Random

PRESETS

Up to 16 presets can be stored and recalled. These presets keep all the sequencer settings: **gate, trigger and slide patterns, main mode, tempo, scale, cv assignment, play mode, start, size, clock divider, transpose, duty cycle...**



Saving a preset will freeze the module up to 20 seconds, but loading it is pretty fast. Obviously the position of the faders is not saved into the presets so loading a preset does not mean that you'll get the exact same performance from the module.

After loading or saving a preset, Step Fader mkII will return to the previous *key mode*.

SAVE A PRESET:

click FUNCT -> click SAVE -> click on the desired button where the preset has to be stored

LOAD A PRESET:

click FUNCT -> click LOAD -> click on the button where the preset you want to load was stored

KEYBOARD, RECORD & CLEAR

The 16 illuminated buttons can be used to play the sequencer as a sort of keyboard in the **KEY** mode, but it can also record the played sequence by entering the **REC** mode. In this mode, a sequence of **64 steps** (no matter what the settings for size are) will memorize the steps where a key has been pressed. The **reset input** will force the sequence to start again, so if you need a shorter sequence than 64 steps use it to determine the length.

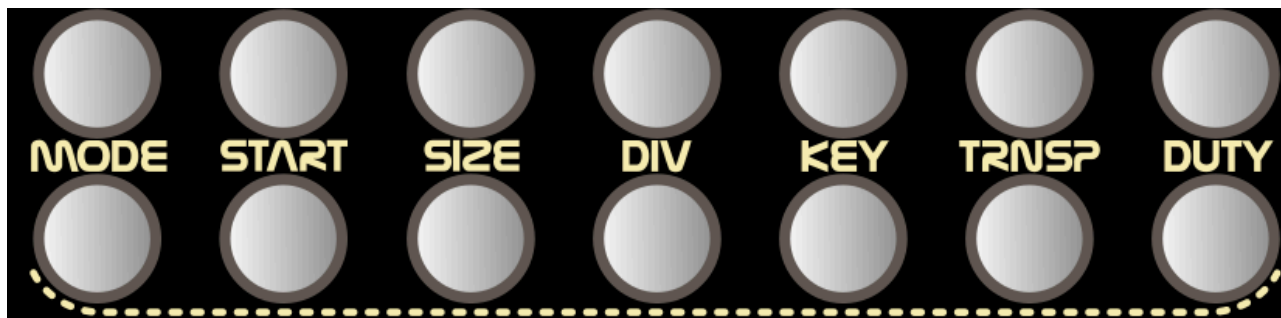
To erase the recorded sequence press **SHIFT + CLEAR**.

If a CV input is assigned to KEY, the play head of that sequencer will be controlled by the cv input and none of the clocks will have any effect on it.



CV INPUT ASSIGNMENT

The sequencer has 2 CV inputs, which can control only one parameter each. Those parameters are the **MODE**, **START**, **SIZE**, **CLOCK DIVIDER**, **KEY** (position), **TRANSPOSE** and **DUTY**. Clicking on the upper row will make that CV input to control a parameter of the first sequencer, and lower row of the second one.



The 2 pots labeled CV IN 1 & CV IN 2 adjust the offset for each of the 2 CV inputs. When nothing is plugged in the CV jacks, those pots will work as manual controls. If a signal is connected to the jack, the respective pot will set the input voltage range:

- CCW / min: voltage range 0v to +10v.
- Center : voltage range from -5v To +5v.
- CW / max: voltage range from -10v to 0v.

Voltages outside those limits will be trimmed / ignored.

So, for example, if you want to use an envelope let the potentiometer CCW, and if you want to use an LFO set the pot to center position.

Please note that if the sequencer *main setup* is set to PARALLEL or SINGLE, the CV assignment for the second sequencer (lower row) will have no effect, except for the duty cycle.

Pressing shift when in CV INPUT ASSIGNMENT mode will display the CV read value using the 2 rows of illuminated buttons as a bar graph..



The **slide time** can also be controlled by CV (it will affect both sequencers) by clicking on the leftmost button of the lower row (the SLIDE one) when in the CV assign mode. The concept is more about adjusting the taste of the slide to your needs, rather than modulating the slide time, which, on the other hand, can be done.

By default CV IN1 (and so its tall trimmer) is assigned to control the size of the sequencer 1, and CV IN2 the clock divider (aka step repeat) of the sequencer 2.

INTERNAL CLOCK & TEMPO CONTROL



Step Fader mkII has an internal clock. Hold SHIFT and click on PLAY to start /stop. SHIFT + RESET will do a manual reset to both sequencers.

The default tempo for the internal clock is 120 BPM. By pressing SHIFT while clicking on - **tempo** + the internal clock will decrease/increase in 2 BPMs. Another way to set the tempo is to click on the TAP button at the desired speed while pressing shift. One tap for every 4 steps.

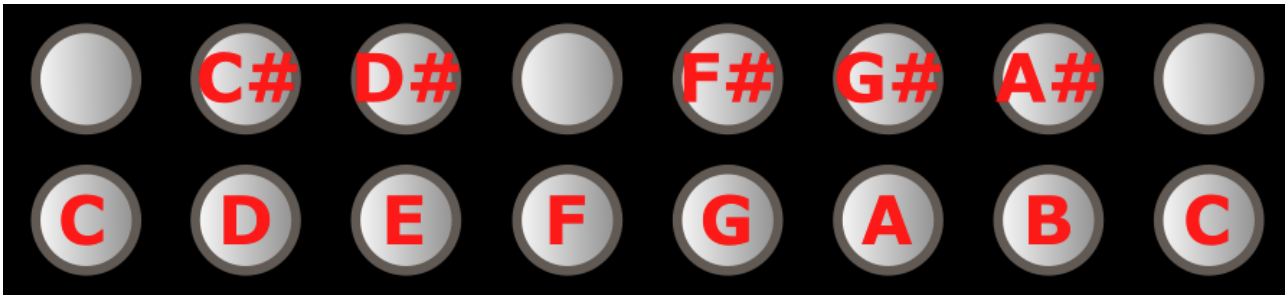


QUANTIZER

Step Fader mkII has a built-in 2 channel quantizer with a range of 0 to +10 volts. These quantizers can be used to process external CV and **they are totally independent of the sequencer except for the slide** (when sequencer 1 is on a step with the slide active the quantizer will perform a slide when a voltage change is received).

Since the inputs of both quantizers have the sequencer outputs normalized to them, a cable from the 'CV 1 + 2' output has to be plugged into either quantizer to get the V/OCT output of both rows when the *main setup* of the sequencer is set to 'SINGLE'. In this setup, all slides (top and bottom rows) will affect only the first quantizer.

The scale of the quantizer can be set using the illuminated buttons by clicking on FUNCT and SCALE, bottom row represents white keys of the piano and top row the black ones (flat/sharp notes)



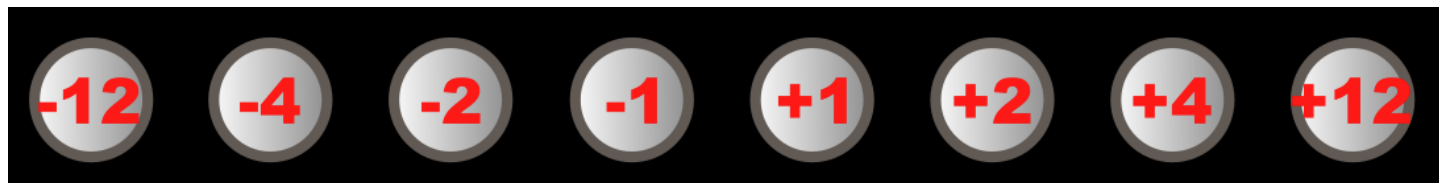
scale examples:

Chromatic	
C major, A minor	
C minor, Eb major	
C minor pentatonic with chormatism between IV and V	
Raga Charukeshi, C mixolydian b6, F melodic	

TRANSCOPE (see also [this](#))

A transposition of the V/OCT output is possible (this will be applied before the quantizer, so the result will always be within the set scale). When in this *key mode*, the upper row of illuminated buttons will set the transposition for the first sequencer, and the lower row will set the second one.

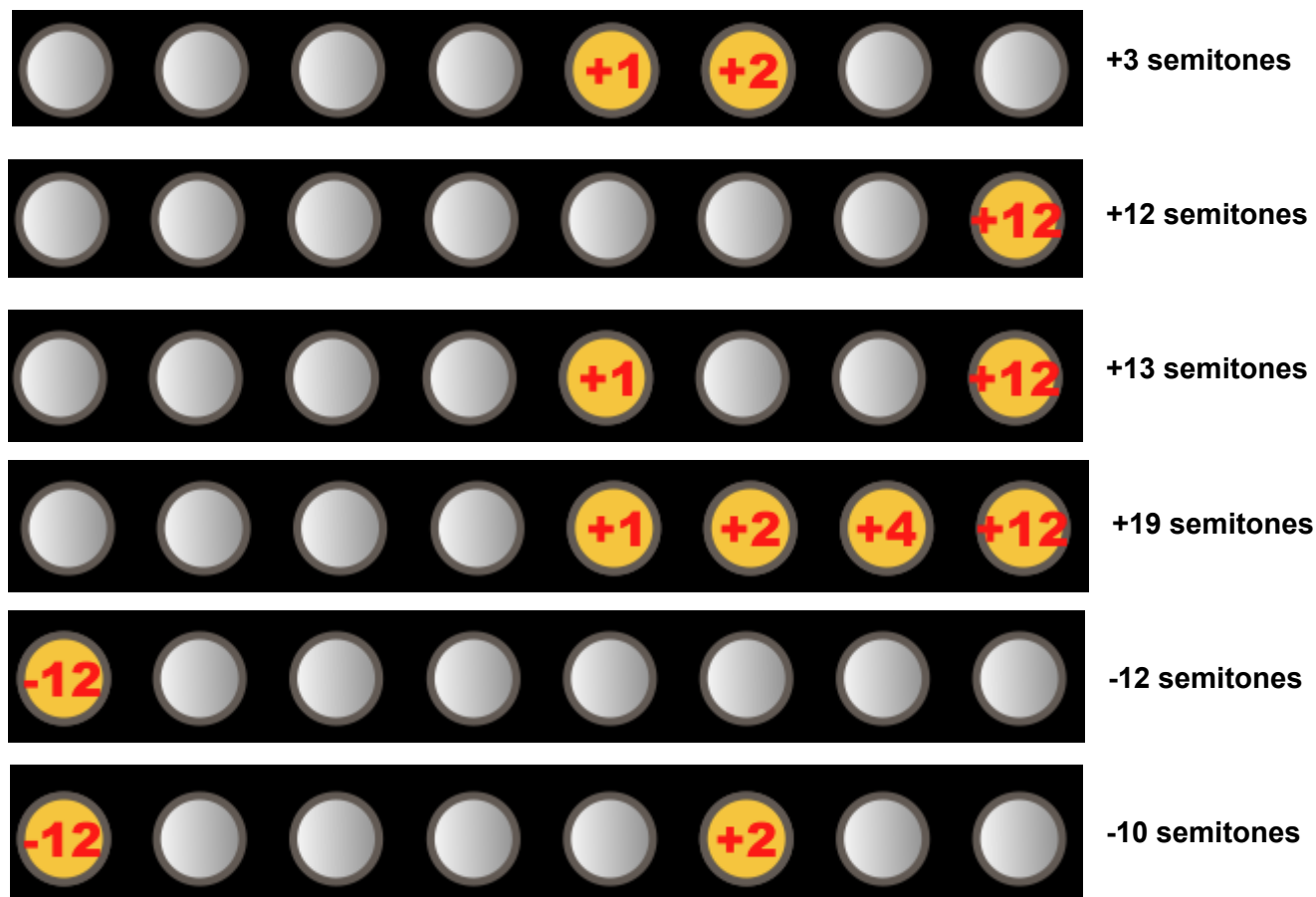
Each button represents an amount of semitones to be added or subtracted as follows:



The transposition will be the sum of all active buttons. For example, to get +3 semitones, set ON the +1 and the +2 buttons.

Setting all buttons ON will have the same result as all buttons OFF, so basically, there are several ways to get the same result. This might seem confusing at first glance, but is a versatile way to manage a range of ± 19 semitones with only 8 buttons. Plus, transposing one octave up or down is pretty easy (just click on the -12 or +12).

Here are some interval examples:



DUTY

As said before, this parameter sets the duration of the triggers (from 5 milliseconds to $\frac{7}{8}$ of the step length) as follows:

5 ms.	1/8	2/8	3/8	4/8	5/8	6/8	7/8
-------	-----	-----	-----	-----	-----	-----	-----

SLIDE AND SLEW LIMITER

These are two different things, the **slew limiter** it's an analog circuit that will continuously smooth the voltage changes present at the slew input (by default the voltage coming from the sequencers), it has no voltage control, while the **slide** will perform a glide when it's active in a step. The slide time can be modified and even controlled by CV.

AVAILABLE TRANSPOSITION METHODS

There are two methods to apply transposition.

First one is the **TRANSPOSE** parameter, that will affect the quantizers only and can be controlled by CV, please note that the minimum voltage for the V/OCT outs is 0V, so negative transposition may not be performed in the lower notes.

Second method is to use the sum output (CV 1+2), for example, if it's connected to one of the quantizers input, the result will be like one sequencer transposing the other, also a voltage can be plugged into one of the slew inputs to perform that transposition with that external signal.

ADDITIONAL OUTPUTS

In the back of the module there is an 8 pin header with some additional things, they can be connected to a jack directly with no additional components:

1 & 2 are clock outputs. If a front panel clock out is set to perform the clock division then the additional one won't do it ...and vice versa.

3 is a random gate based on the clock of the sequencer 1.

A & B are the voltages coming from the sequencers (A for the upper row, B for the lower one). Let's say quantizer 1 and slew 1 are processing external voltages ...in such case, none of the front panel outputs would be providing the sequencer 1 voltage, so there you have it.

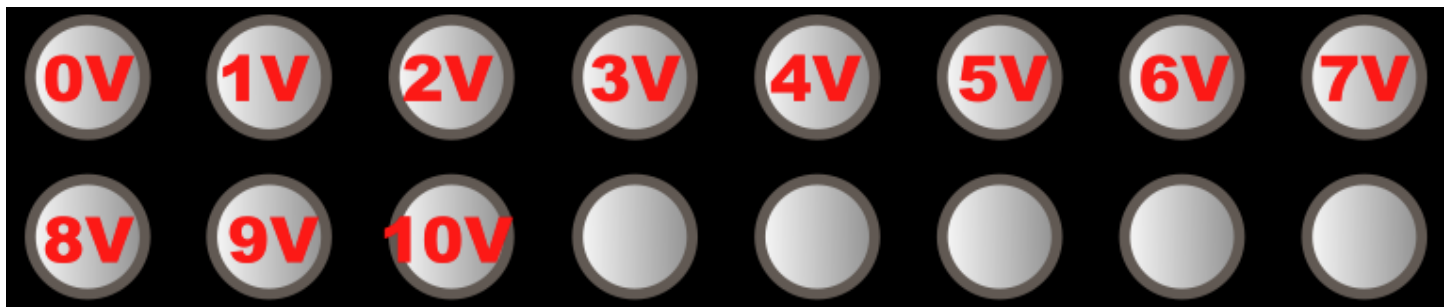
The rest of the pins are ground.

V/OCT CALIBRATION

Assembled units are calibrated already so this step is not necessary, but if you want to calibrate it can always enter the calibration mode by holding shift while powering the module until the first illuminated button starts pulsing. I don't recommend doing the calibration unless you are absolutely sure that you want/need to do it.

The process is simple:

- Unplug any cable from the CV IN 1 and CV IN 2.
- Power the module, hold shift until the first button shows a pulse.
- Connect a multimeter to the VOCT output.
- The first illuminated button will pulse, that means 0 volts. Use the CV IN 1 & 2 pots to adjust both V/OCT outputs respectively so you get a value as close to 0 volts as possible.
- Use the illuminated buttons to set other output voltages (the module needs 11 calibration points: 0V, +1V, +2V...+9V and +10V).
- Use the CV IN 1 and CV IN 2 tall trimmers to adjust the output voltage as close as possible for every calibration point and V/OCT output (it will change in steps of about 3 millivolts, so for 1 volt 1,001v or 0,999v would a valid value).
- Do not press another button until both outs are calibrated. If some steps have been calibrated and you go back to any of them, then they have to be calibrated again.
- **V/OCT INPUTS CALIBRATION:** connect 2 voltage sources to QIN1 and QIN2. Input 0 volts and click on the button 15, input +3 volts and click on button 16.
- To save the settings, press the FUNCT button for about 1second, this will freeze the module for several seconds, DO NOT TURN THE POWER OFF...just wait until settings has been saved. To exit the calibration mode without saving the settings, first click on the FUNCT button and then any of the illuminated buttons to enter any other key mode.

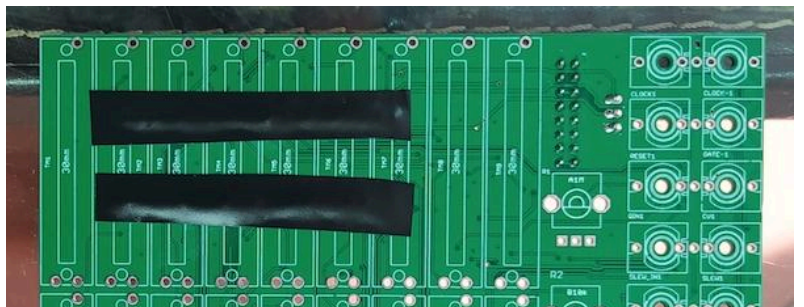


ASSEMBLY GUIDE

1. Cut the pins of the female headers for the MCU board, (ideally to 1,6mm), the best way for this is to use the PCB as a guide.



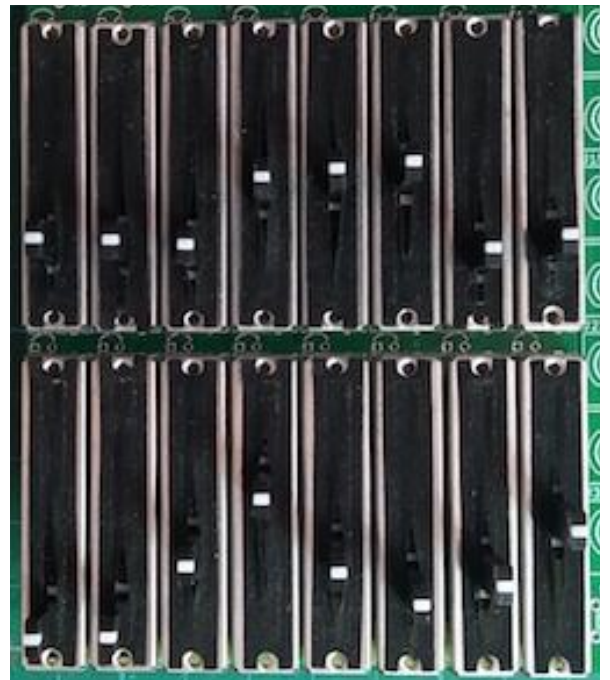
2. Solder the female headers for the MCU board and put some insulating tape to avoid shorts with the faders.



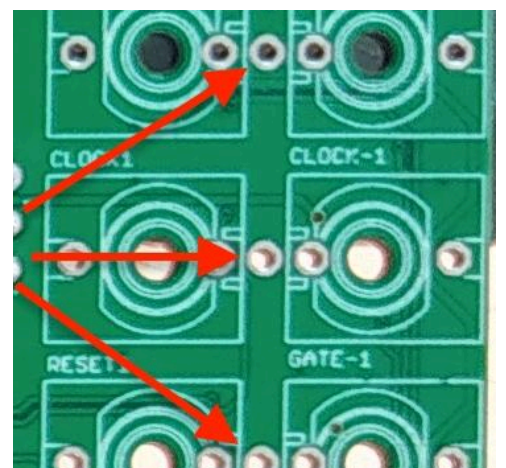
3. Next solder the faders, take care to **not apply too much heat (max 340° during 3 seconds)**, otherwise, faders could be damaged.

4. Solder all the other headers & header box, **mind the orientation of the header box**, which is indicated by the soldermask in the PCB.

5. Place the tall trimmers (for B10k ones a value of B100k would be fine too).

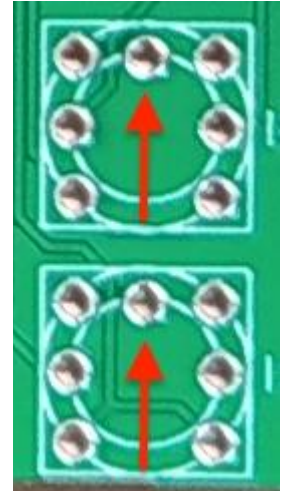
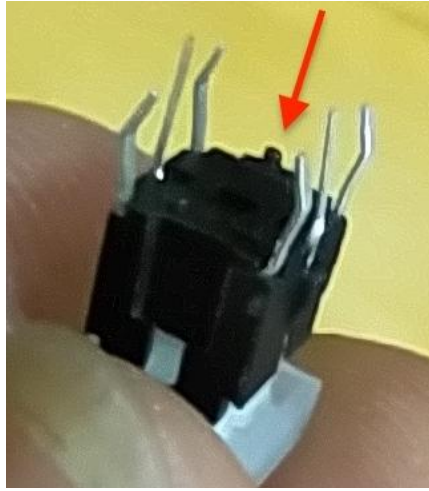


6. Place the thonkiconn jacks, please note that every pair shares the same hole for the ground pin.



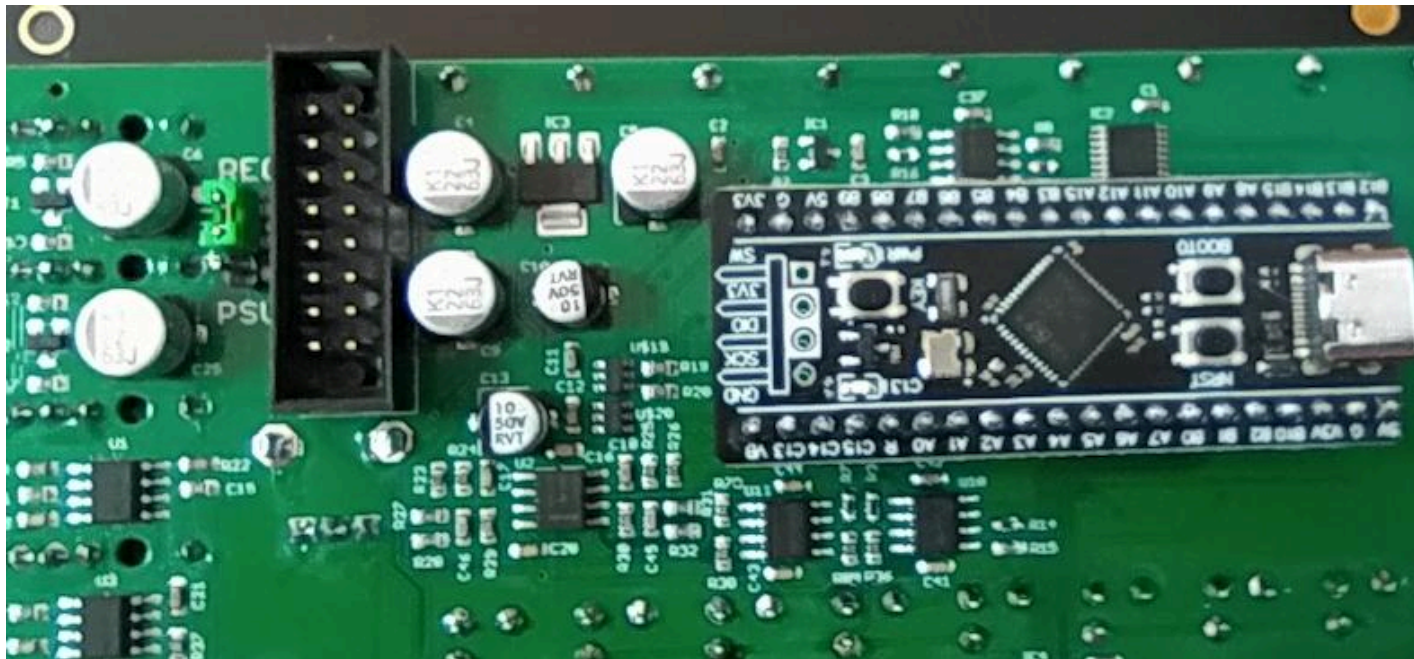
7. Carefully place the push buttons. The orientation of the illuminated ones is indicated by the 'plastic leg', which goes in the top-center hole. Put special attention so all legs are correctly inserted in their respective holes.

8. Place 4 of the nylon screws on the panel and secure them with a nut on the other side, **do not over-tighten the nut so that the screw can turn.**



9. Place the panel and put all the nuts before soldering. Turn the screws just a little so the panel is parallel to the PCB.

10. Put the jumper in the 1x3 pin header in REG position. Place the MCU board and proceed with the [calibration](#).



<https://www.tesseractmodular.com/>

<https://www.modulargrid.net/e/vendors/view/713>

<https://www.facebook.com/Tesseract-Modular-146727136274298>

https://www.instagram.com/tesseract_modular/